Microwave reheating of electron-beam-pumped rare-gas–halide laser plasmas: a robust observable, a real but marginal channel

Cooper K. Watson

Texas A&M University[TODO: verify department/affiliation line]

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Abstract

Whether microwave energy added to an electron-beam-pumped KrF/ArF laser plasma can improve its performance is assessed *in silico* across three independently constructed electron treatments: a 26-species zero-dimensional kinetics engine with interchangeable electron-rate tiers (LoKI-MC swarm tables and rates measured by an in-house PIC/MCC code), and an independently developed one-dimensional discharge-pumped fluid model with an in-tree two-term Boltzmann solver. Using twin-pair numerical experiments (identical runs with and without a prescribed-power microwave window) over eight benchmark configurations, we find a three-part answer. (i) Reheating is a *robust observable*: the gain-proportional sidelight fluorescence rises in every benchmark on both rate tiers and in the independent model (+1.5% to +12%), so hybrid-pumping experiments that monitor fluorescence will genuinely detect the microwave. (ii) One laser-energy-positive channel is *real and is not an artifact*: afterglow reheating of a high-optical-margin cell that is still extracting raises laser output by $\sim +10\%$ per microwave joule, and this sign survives all 41 rate-uncertainty and literature-conflict bracket variants (+8.3% to +12.9%) and a change of electron-rate tier in a sibling configuration — and it is reproduced at the same $\sim +10\%$ scale in the independent one-dimensional model, where its predicted margin, window-timing, and deposition-locality dependences are verified ladder by ladder under timestep refinement. (iii) Exploiting it is *very difficult*: at a program-representative coupling budget (250 kW per litre) no benchmark changes its blended intrinsic efficiency by more than 0.006 percentage points; the conductive plasma pins the settled reduced field at $\lesssim$2 Td at any injected power, both excimer supply chains weaken when the electrons are heated, the wall-plug inequality $\eta_{\text{source}} \times m > \eta_{\text{beam}} \approx 7\%$ is barely reachable even by magnetron-class sources, and the sign of the during-pulse variant is hostage to a single rate coefficient (positive only if $e + \text{Kr}^* \rightarrow \text{Kr}^+ + 2e$ lies within a factor $\times 0.49$–$\times 1.99$ of its nominal value). We state what would decide each remaining open question: a factor-two determination of that one rate, and gain-referenced twin-shot experiments whose absorbed-energy denominator is closed by a full-wave deposition map.

I. INTRODUCTION

Electron-beam-pumped rare-gas–halide lasers (KrF at 248.5 nm, ArF at 193 nm) remain the leading broadband-ultraviolet driver candidates for inertial fusion energy, and their

intrinsic efficiency — of order 7–10 % in the best demonstrated devices [1][TODO: verify] — is the quantity every driver study tries to raise. A recurring proposal is *hybrid pumping*: keep the electron beam as the primary ionizer and add electromagnetic power (microwave-band, waveguide-coupled) to reheat the electron population, either during the pump pulse or in the recombining afterglow, in the hope of harvesting additional excimer formation from the ion–ion and harpoon chains [2][TODO: verify or drop].

The proposal is attractive because the microwave subsystem is cheap relative to the beam, and because experiments monitor a convenient surrogate: the sidelight fluorescence at the lasing wavelength, which is proportional to the small-signal gain. It is also easy to test badly — a fluorescence increase does not by itself demonstrate an efficiency increase, and a simulated efficiency increase can be an artifact of a single uncertain rate coefficient, of a solver tolerance, or of a timestep. This article is a falsification-first computational study designed to separate those outcomes. We ask three sharply distinguishable questions: Is reheating *observable*? Is any laser-energy gain *real* in the sense of surviving every model-side perturbation we can apply? And what does it *cost* to exploit?

Our contribution is methodological as much as physical. Every claimed sign is subjected to a decidability ladder — solver and integrator convergence, an exchange of the complete electron-rate tier between independently produced sets, brackets of $\times\frac{1}{3}.. \times 3$ on every estimated rate plus literature-conflict alternates, and replication in an independently written model of a different pumping geometry — and any sign that fails a rung is reported as *undecided*, together with the measurement that would decide it.

II. MODELS AND METHODS

A. Three electron treatments

(a) Zero-dimensional kinetics engine ("REBUILD"). A 26-species KrF/ArF plasma-chemistry engine (electrons, electron energy, gas temperature, cavity photon field, and a closed four-channel energy ledger integrated as state) with ∼169 reactions in the KrF configuration. Beam deposition uses Spencer–Fano-partitioned yields; the laser field is a mean-field cavity with an intracavity absorber inventory. The engine is predictive-only: literature anchors are scored against, never fitted to. Two interchangeable electron-rate tiers supply

every electron-impact rate: (i) a LoKI-MC swarm-table tier [3][TODO: verify LoKI-MC citation] keyed by mean electron energy, with measured relaxation-bridged curves below table coverage, and (ii) a *kinetic* tier of rates measured by an in-house 3-D PIC/MCC code from the same cross-section sets. Agreement between tiers is diagnostic, not enforced.

(b) PIC/MCC rate source. The kinetic tier's rates are exported from particle simulations (Monte Carlo collisions on the full cross-section set) as rate-versus-mean-energy curves; they share nothing with the swarm tier but the cross sections.

(c) Independent 1-D discharge model. A one-dimensional charged-fluid model of a discharge-pumped KrF laser (Scharfetter–Gummel transport, Poisson field, magnetic-pulse-compression circuit, 45-channel reduced kinetics, in-tree two-term Boltzmann solver refreshed from the local mean energy), developed independently by another author [4][TODO: citation/permission for the 1-D model]. We drive it unmodified through a subclass that adds a bounded prescribed-power heating term at the model's own deliverable-heating seam; the microwave energy is metered externally.

B. Twin-pair experiments and metrics

Every reheat question is posed as a *twin pair*: two runs identical in every input except that one carries a trapezoidal microwave window of prescribed *injected power* P_{MW} (power, not field: a prescribed field across a conducting plasma is an unbounded generator — in an early design study a held 8 Td window drove a cell to full ionization while the energy ledger closed perfectly; the bookkeeping was right and the model assumption wrong). In the 0-D engine the field settles each step at the lowest reduced field E/N at which the swarm absorption $e\, n_e\, v_d(E/N)\, E\, \nu^2/(\nu^2+\omega^2)$ matches the prescribed power, capped at 30 Td. Windows: *afterglow* (opening 10 ns after beam end, 150 ns flat) and *co-pump* (the second half of the beam pulse). The coupling budget follows a program-representative rule of 250 kW per litre of pumped volume at 95 GHz (the one X-band configuration retains its own hardware value).

Metrics: the *marginal* $m = \Delta E_L / E_{\mathrm{MW}}$ (laser energy gained per microwave joule delivered); the *blended* intrinsic efficiency $\eta = E_L/(E_{\mathrm{beam}} + E_{\mathrm{MW}})$, whose change $\Delta\eta$ answers the efficiency question with the microwave in the denominator; and the post-beam fluorescence energy $E_{\mathrm{fl}} = \int (n_B/\tau_B)\, h\nu\, V\, dt$ over the window, the experimental observable.

C. The decidability ladder

A sign is reported as *decided* only if it is invariant under: (1) numerics — seven solver settings including an independent integrator (twin differences agree to $z \approx 3.7 \times 10^3$ in the reference case); (2) the electron-rate tier swap; (3) brackets of $\times\frac{1}{3}$ and $\times 3$ on every estimated-provenance rate and every literature-conflict alternate, applied to both twins; and (4) where expressible, replication in the independent 1-D model, including timestep refinement. Anything else is *undecided*, and we compute what would decide it.

III. RESULTS

A. Reheating is a robust observable

Table I collects the eight benchmark configurations: two program-class cells that are sub-threshold in this engine, the program's high-power cell, an X-band-equipped chamber column, two optimized operating points found by a lever scan (Ar- and Ne-buffered), and two literature anchor cells (a 1985 KrF kinetics benchmark and a large ArF device at its recorded peak-efficiency point) [1, 5][TODO: verify].

The post-beam fluorescence rises with the microwave on in *all eight* cases, on both rate tiers, spanning +0.07 to +15 mJ over the window at the stated coupling rule. The independent 1-D model agrees: at its lasing point the afterglow window raises its fluorescence proxy by +11.8%. This is the answer to the observability question, and it carries a warning for experiment design: a hybrid-pumping campaign that scores the microwave by gain-proportional fluorescence will see a positive response *even in configurations where the laser output falls* (six of eight afterglow cases in Table I). The fluorescence statement and the efficiency statement are different statements.

B. One laser-positive channel is real — and it is not an artifact

Exactly one benchmark gains laser energy from afterglow reheating: the Ne-buffered optimized cell, whose marginal is +10.0% ($\Delta\eta = +0.0024$ pp). Three independent tests argue this is physics, not numerics or a rate accident.

Bracket robustness. Re-running the twin pair under every estimated-rate bracket ($\times\frac{1}{3}$ and

TABLE I. Twin-pair results across the benchmark set (swarm tier, corrected engine; coupling rule 250 kW/litre). Marginal $m = \Delta E_L / E_{\mathrm{MW}}$; $\Delta\eta$ is the blended-efficiency change in percentage points (microwave joules in the denominator); ΔE_{fl} is the post-beam fluorescence change for the afterglow window. The two sub-threshold cells lase in neither twin.

		afterglow window			co-pump window	
case	η_0 (%)	m (%)	$\Delta\eta$ (pp)	ΔE_{fl} (mJ)	m (%)	$\Delta\eta$ (pp)
program class, 3 atm (Ar)	no lasing	—	—	+0.07	—	—
program class, 3 atm (Ne)	no lasing	—	—	+0.18	—	—
high-power cell, 2 atm	1.54	−2.8	−0.0055	+0.48	+12.5	+0.0029
X-band chamber column	2.16	−10.4	−0.0036	+2.15	−1.9	−0.0004
Ar-optimized point	4.95	−2.1	−0.0056	+5.12	+1.3	−0.0010
Ne-optimized point	6.19	**+10.0**	**+0.0024**	+1.30	+5.1	−0.0002
KrF anchor cell (1985)	0.70	−0.0	−0.0003	+0.47	+3.0	+0.0002
ArF anchor, 1.09 atm	6.13	−1.3	−0.0031	+14.8	+5.7	−0.0001

×3) and every literature-conflict alternate — 41 variants — leaves the marginal positive in *all 41*, spanning +8.3% to +12.9% [Fig. 1(a)]. Notably, the rate that controls the co-pump sign (Sec. III C) moves this marginal by less than one percentage point.

Cross-tier and cross-model witnesses. The same high-margin-afterglow class appears on the independent kinetic rate tier in the sibling Ar-optimized cell (−2.1% on the swarm tier, +9.5% kinetic), and in the independent 1-D discharge model, whose afterglow marginal +0.89% → +1.06% is stable under timestep halving. No single case is yet confirmed positive on *both* tiers — the Ne-optimized kinetic-tier run is computationally severe and still integrating — so we state the class-level claim only.

Mechanism. A row-flux audit of every excimer-formation channel shows why this case is special and the others are not: all direct formation rows are heavy-particle, and heating the electrons *weakens both supply lines* (the F^- fuel of the ion–ion rows via attachment, ×0.13 from 0.5 to 3 eV; the metastable fuel of the harpoon rows via dissociative-recombination throughput, ×0.37). The gain in the one positive case is not afterglow harvest at all — it is *late extraction*: the high-margin cavity is still draining while the window runs ($\Delta E_L \gg \Delta E_{\mathrm{fl}}$ there), i.e. a late co-pump of a still-lasing cell. The channel is real, small, and conditional

on optical margin.

One-dimensional verification. If the gain is late extraction, three predictions follow: it must grow with the optical margin, die as the window slides past the cavity drain, and depend on *when* far more than on *where* the power is deposited across the gap. All three hold in the independent 1-D model [Fig. 2]. Scanning the output coupler at fixed drive, the marginal rises monotonically (+0.01%, +0.20%, +0.78%, +0.9% for $r_{\mathrm{out}} = 0.3$–0.9) even though the *absolute* laser output peaks at $r_{\mathrm{out}} = 0.6$: the marginal follows the photon lifetime — the drain-overlap knob — not the laser performance. At the high-margin point, extending the integration horizon to capture the full drain raises the marginal to +5.7% (+10.1% at half timestep; robustly positive, magnitude unconverged upward) — quantitatively the same $\sim +10\%$ class as the 0-D channel. Sliding the same window later collapses it: +1–3% at partial overlap once timestep-refined (the coarse-step −4.4% there is another discretization artifact, resolved by $\Delta t/2$ and $\Delta t/4$ repeats), and $-0.2\% \approx 0$ past the drain. Finally, re-depositing the same power as a 2-mm edge skin, a center-weighted profile, or uniformly moves the marginal by at most ∼0.3 pp, with edge and center indistinguishable to 0.04 pp: across the gap, *where* the power lands is second-order and *when* it lands is first-order. The channel and its mechanism survive spatial structure.

A placebo-calibrated significance test. Finally, we close the artifact question as a formal hypothesis test. Per solver setting (seven, spanning two tolerance decades, tightened absolute tolerances, a capped step, and the independent Radau integrator), the 0-D twin pair is run three ways: off, at the 500-kW signal, and with a 1-W *placebo* window whose twin difference — after subtracting the first-order physical response — calibrates the numerical differencing noise in situ. Modeling the signal and placebo ensembles with a common profiled noise scale, the artifact-only null is rejected with exact one-sided $p_0 = 1.7 \times 10^{-61}$ (Student-t, 13 dof), i.e. $Z = 16.5\sigma$; 2×10^5 null toys give zero exceedances; the 95% $\mathrm{CL_s}$-allowed signal band is [8.5980, 8.5982] mJ, excluding zero by 1.4×10^5 standard errors; the worst of the 41 bracket variants alone stands at $z \approx 4 \times 10^4$; and the calibrated noise would have to be underestimated by a factor $\sim 4 \times 10^4$ to reduce the discovery to 3σ. The calibration is itself conservative: the placebo difference contains a real small-signal response (the response is strongly sublinear in power), which inflates the noise estimate roughly forty-fold above the raw solver scatter, and a split-sample negative control behaves as required. In the 1-D model the deterministic differencing floor, measured by a power ladder down to 250 W,

is 8×10^{-4} mJ — a factor ∼300 below the signal. These statistics reject a numerics-noise origin of the effect; by construction they cannot certify against a conceptual error shared by a physics model itself — that burden is carried by the independent-implementation agreement, the closed energy ledger, and the bracket program of Sec. II.

C. Why it is very difficult

Five quantitative walls separate "real" from "useful."

(1) The efficiency ceiling at feasible coupling. Across all eight benchmarks and both tiers, no blended efficiency moves by more than 0.0056 percentage points (kinetic tier: 0.0036). The positive marginal must exceed the cell's own efficiency for the microwave joules in the denominator to pay; even the +10% marginal of Sec. III B barely clears its cell's 6.19%, and the co-pump marginals that exceed their cells' η_0 do so by amounts worth $\lesssim$0.003 pp at the stated coupling.

(2) The conductivity pinning. Prescribing more afterglow power does not reach the reduced fields where electron-impact formation pays ($E/N \gtrsim$ 20–30 Td): the conductive plasma settles at $\lesssim$2 Td at any injected power (probed to 1 GW), because added power raises n_e faster than E. The one heat-positive route (direct excitation, whose rate rises by eight orders of magnitude from 0.5 eV to the 30-Td swarm point) is therefore unreachable in the afterglow.

(3) The supply-chain inversion. As measured by the audit above, both feedstocks of excimer formation fall when the electrons are heated; the identical audit of the ArF chain (six direct formation rows, all heavy-particle) returns the same verdict with slightly worse positioning. Electron heating pays in laser families whose upper level is pumped by electron impact at eV energies (CO_2/CO class); the excimer chain is the wrong class.

(4) The wall-plug inequality and the frequency question. With collision rates $\nu_m = 0.9$–$8.9 \times 10^{12}\,\mathrm{s}^{-1}$ over the operating band, the collisional/field-oscillation crossover sits near 0.3 THz: below it the plasma cannot distinguish 1 GHz from 300 GHz (equal absorption lengths, 1.8–2.0 mm at beam-on electron density), and above it coupling collapses as ω^{-2}. Wall-plug viability requires $\eta_{\mathrm{source}} \times m > \eta_{\mathrm{beam}} \approx 7\%$: Fig. 1(b) shows only magnetron-class sources times the *most favorable and least robust* marginal (+12.5%) barely clear it (8.4–8.6%); millimeter-wave gyrotrons (4.2%) and CO_2 lasers (1.4%) do not. The same 2-mm

absorption length implies the deposition is a moving skin in any cell larger than centimeters — the uniform-coupling assumption of every model here is an upper bound that real launch geometry can only degrade.

(5) The during-pulse sign is one uncertain rate wide. The co-pump marginal at the high-power cell's plateau is +8.0% nominally, and *every* bracket variant but one leaves it positive. The exception owns the sign: stepwise ionization of the krypton metastable, $e + \mathrm{Kr}^* \rightarrow \mathrm{Kr}^+ + 2e$. Mapping the marginal against that single multiplier [Fig. 3(a)] gives a sign-positive window of $\times 0.49$–$\times 1.99$ about the nominal rate — geometrically centered on it, with two zero crossings inside the $\times 3$ uncertainty band appropriate to the rate's estimated provenance. The sign is therefore *undecidable at current rate knowledge, and decidable by a factor-two determination of one rate.* Consistently, the independent kinetic tier (a different effective rate set) returns −2.7% to −4.5% at the same nominal point, and the independent 1-D model's co-pump marginal collapses from +16.6% at 50 ps to −24.8% and −5.8% at 25 and 12.5 ps [Fig. 3(b)]: the coarse-timestep positive is a discretization artifact, and the fine-timestep values are small and negative while the baseline is still converging.

IV. DISCUSSION

What the study licenses an experimentalist to claim. A microwave window *will* produce a measurable, model-robust fluorescence response in essentially any e-beam-pumped configuration — reheating is feasible as an observable and as plasma physics. A laser-energy gain from afterglow reheating exists in a specific, predictable regime — a cavity still extracting through the window — at the +10%-per-microwave-joule level, and our bracket analysis says a failure to observe it there would falsify the model rather than the instrument. What no configuration supports is a material improvement of blended efficiency at feasible coupling: the levers that actually move these devices are conventional (halogen fraction, pressure, optics, buffer choice — a measured ladder from 1.54% to 6.19% in this engine, roughly 10^3 times any electromagnetic effect found here).

What would decide the open questions. (i) A factor-two determination of the Kr^* stepwise-ionization rate — by literature adjudication of the metastable ionization cross section, by a driven relaxation measurement in the PIC code, or by experiment — decides the during-pulse sign within the swarm framework. (ii) Completion of the kinetic-tier run of the Ne-optimized

afterglow case decides whether the one robust positive is also tier-stable in the same cell. (iii) On the bench, gain-referenced twin shots (window on/off at fixed fill) in a high-margin cell, with the absorbed microwave energy closed by a full-wave deposition map rather than by forward power, measure the marginal directly, and scanning the window delay traces the drain curve of Fig. 2(b) — the mechanism's most direct experimental signature; shot counts for stated significance levels follow from the twin-difference variances measured here.

Limitations. The 0-D engine cannot express transverse structure (filamentation and arc instability are the historical failure mode of during-discharge heating, and are invisible here); the 1-D model resolves the gap direction only, though it suffices to verify the channel's margin, timing, and deposition-locality behavior (Sec. III B). Both microwave treatments prescribe the absorbed power; the 1-D profile ladder bounds the deposition-locality error (second-order), but launch-geometry effects proper — reflection, standing waves, the measured 2-mm skin against a real vessel — remain outside both models. The reduced kinetics sets, while literature-anchored, carry the bracket uncertainties used above by construction. All results are predictions; no reheat experiment has yet been scored against them.

V. CONCLUSION

In silico, microwave reheating of e-beam-pumped excimer plasmas is (i) a robust, cross-model observable through gain-proportional fluorescence; (ii) genuinely laser-positive in one narrow, mechanistically understood regime — late extraction from a high-margin cavity — with a sign that survives every rate bracket, solver, and model-family perturbation applied; and (iii) very difficult to convert into device efficiency: blended gains are capped at the 10^{-3}-percentage-point scale at realistic coupling, the afterglow plasma self-limits the attainable field, the excimer supply chemistry weakens under electron heating, the wall-plug inequality is at best marginal, and the one potentially larger channel (during-pulse co-pumping) has a sign that current rate knowledge cannot fix. The constructive outputs are precise: one rate to measure to a factor of two, one regime in which to run twin-shot tests, and one observable — fluorescence — whose positive response must not be mistaken for efficiency.

ACKNOWLEDGMENTS

[TODO: Insert program-approved funding and collaboration acknowledgments verbatim; do not paraphrase. Acknowledge the author of the independent 1-D discharge model by name with permission.]

DATA AVAILABILITY

All twin-pair results, bracket tables, the rate-multiplier map, and the frequency scan are recorded as JSON artifacts alongside the engines' changelogs in the project workspace; the figures in this draft are regenerated from those artifacts by the included script (`make_figs.py`). [TODO: public archive/DOI before submission]

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[1] [TODO: Electra KrF/ArF program reference (NRL); e.g. Wolford et al., and the recorded ArF peak-efficiency point used as the anchor.]

[2] [TODO: hybrid e-beam + microwave pumping concept reference, or drop the sentence.]

[3] [TODO: LoKI-MC citation: Tejero-del-Caz et al., Plasma Sources Sci. Technol. (LisbOn KInetics); verify which of LoKI-B / LoKI-MC papers applies.]

[4] [TODO: the independent 1-D KrF discharge fluid model — cite the author's code/paper with permission.]

[5] [TODO: Kannari et al. KrF kinetics benchmark (1985) used as the anchor cell.]

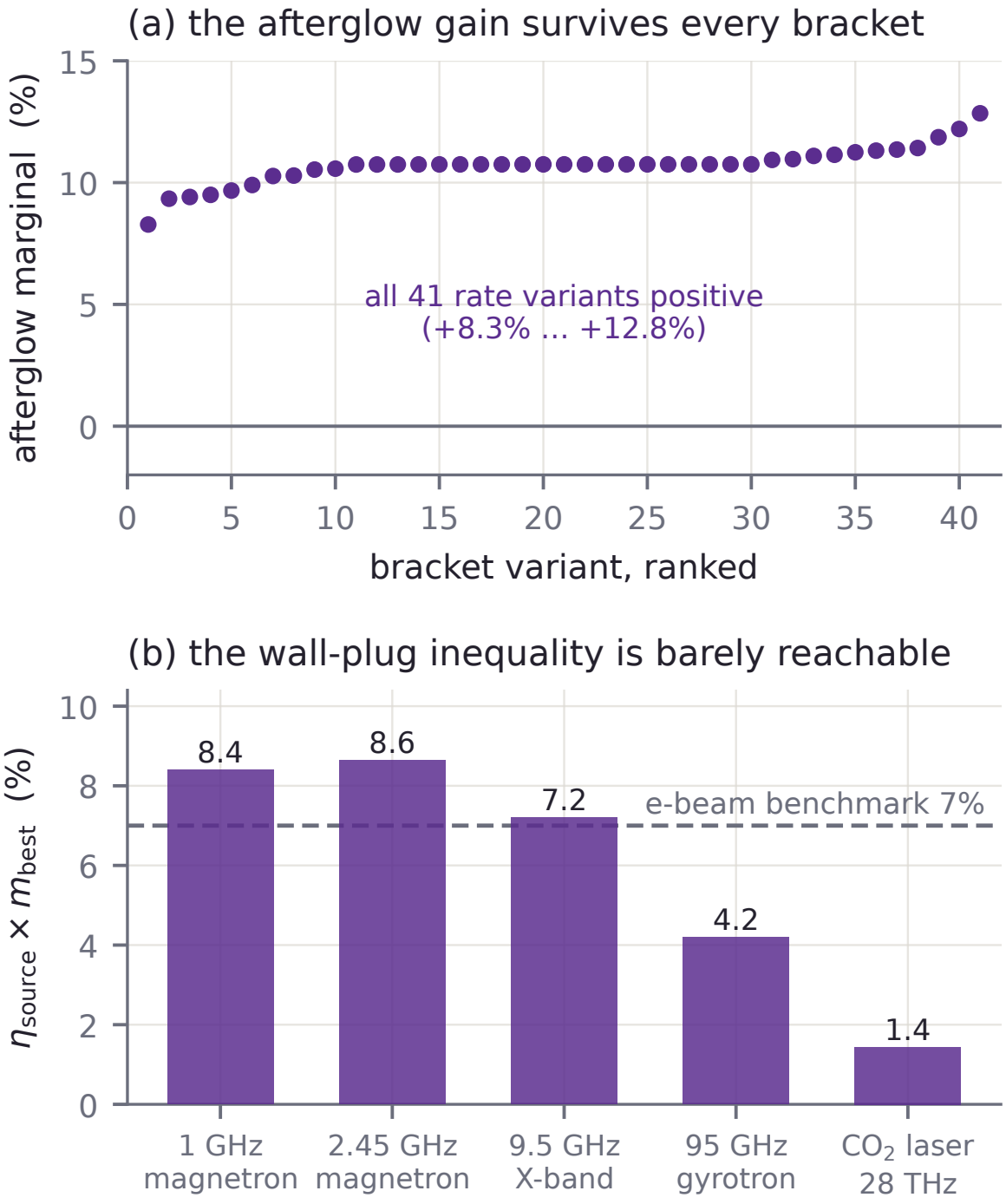


FIG. 1. (a) The one laser-positive afterglow case (Ne-optimized cell) under all 41 rate-bracket and literature-conflict variants: every variant stays positive (+8.3% to +12.9%). (b) The wall-plug inequality: source efficiency times the best (and fragile) marginal, by band, against the ≈7% e-beam benchmark. Data: recorded run artifacts; figures regenerated by the included script.

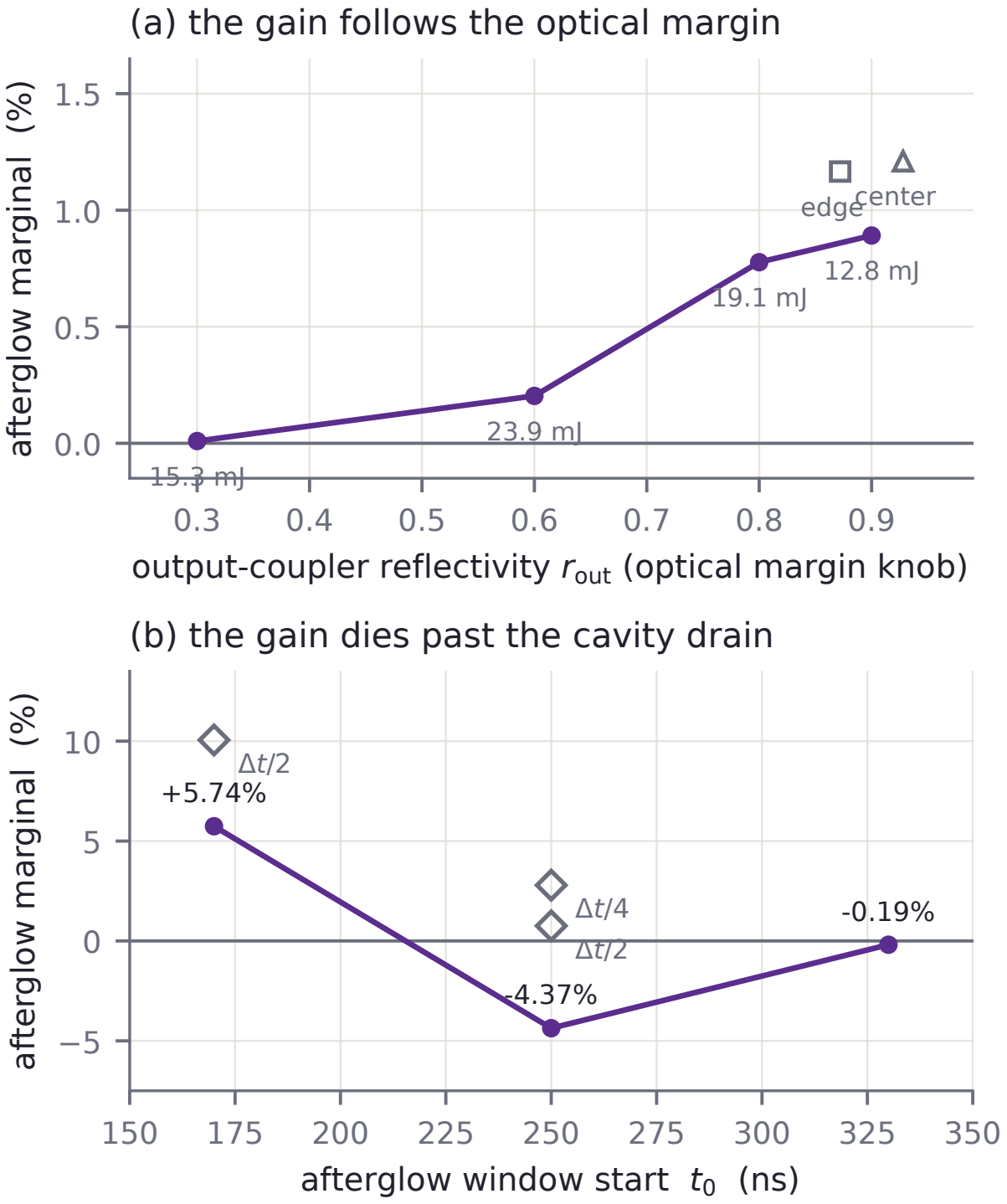


FIG. 2. One-dimensional verification of the late-extraction mechanism (independent discharge model, afterglow window, absorbed power fixed at 250 kW). (a) Marginal versus output-coupler reflectivity (300-ns horizon); the annotated baseline outputs are non-monotone while the marginal is monotone — it tracks photon lifetime, not laser performance. Open markers: edge- and center-weighted deposition variants at $r_{out} = 0.9$. (b) Marginal versus window start at the high-margin point (500-ns horizon, full drain captured); open diamonds are timestep-refined repeats ($\Delta t/2$, $\Delta t/4$).

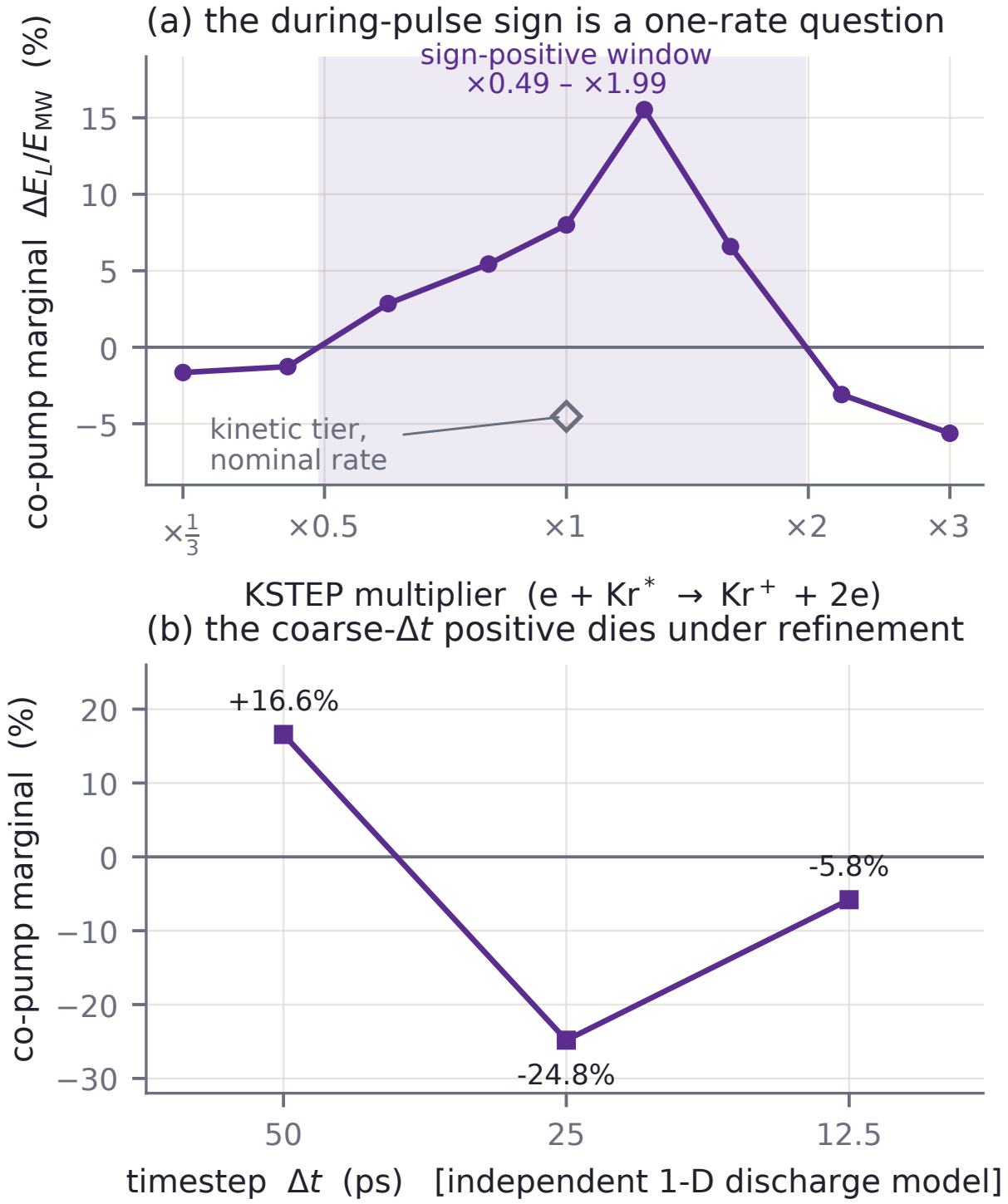


FIG. 3. (a) Co-pump marginal at the high-power cell's lasing plateau versus the multiplier on the one sign-controlling rate ($e + \mathrm{Kr}^* \to \mathrm{Kr}^+ + 2e$), swarm tier. The sign is positive only inside ×0.49–×1.99 (shaded); the open diamond is the independent kinetic tier at the nominal rate. (b) The independent 1-D discharge model's co-pump marginal under timestep refinement.